

Question 3 Class Result Processing. Date:

1) PAC Chart

* Given Data

Total Marks = 100

Subjects = 5

$\geq 80 \rightarrow$ Distinction

* Processing required

Average = $\text{Sum} / 5$

* Required Results

⊗ ~~Print~~ display "Distinction",
"Pass" "Fail" or

"Subject deficiency"

* Solution alternatives

Could use while or
repeat until loop.

2) IPO Chart

Input

Process

Output

* Number of
students, N

* Subject marks
out of 100

* Total all
subjects

* Average = $\text{Sum} / 5$

* Avg $\geq 80 \rightarrow$ Distinction

* Avg $\geq 60, < 80 \rightarrow$ Pass

* Avg $< 60 \rightarrow$ Fail

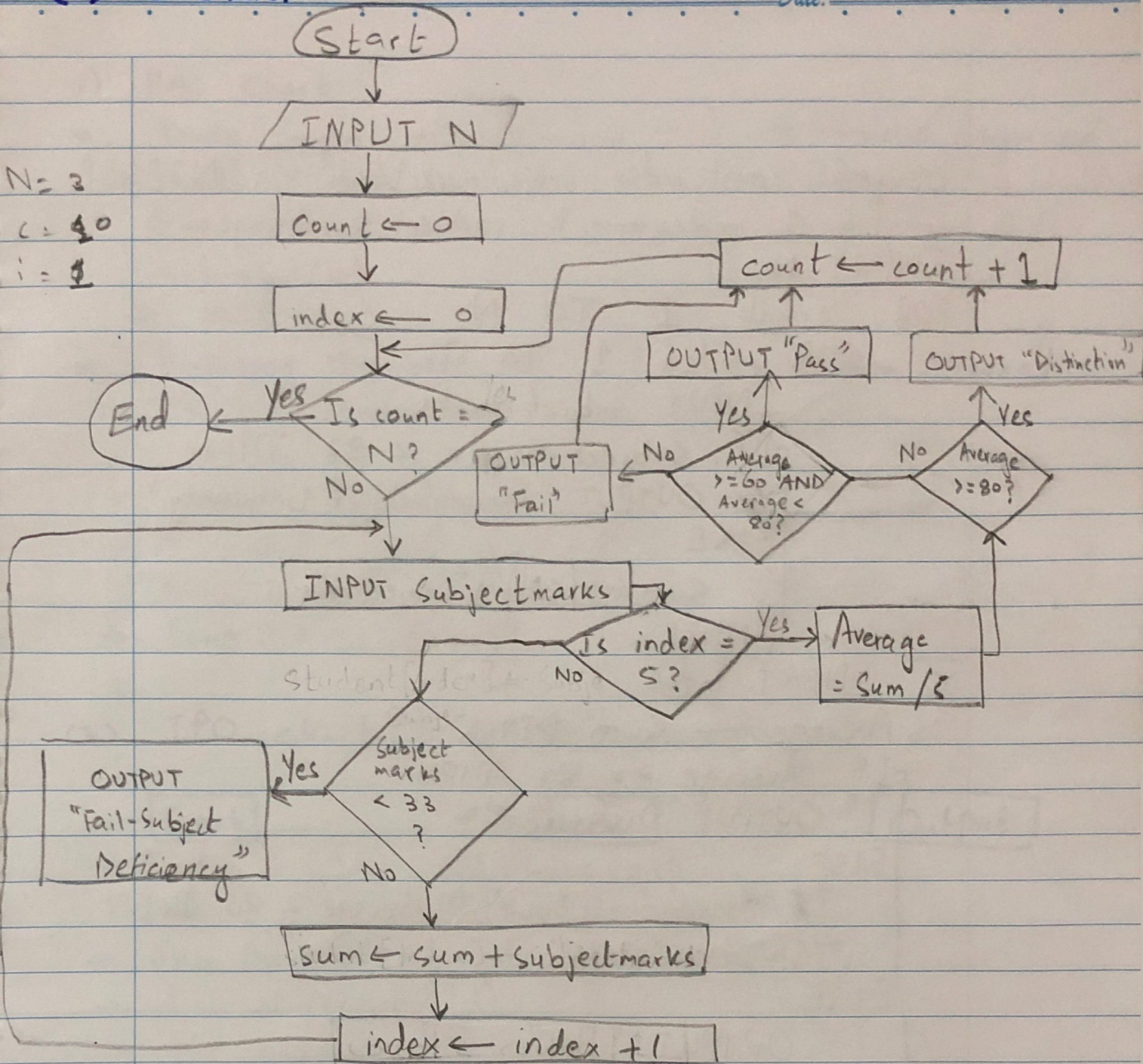
(3) Flowchart.

Date: _____

$N = 3$

$c = 0$

$i = 1$



(4) Pseudocode.

Date: _____

```
DECLARE Average : REAL
DECLARE count, index, subject marks, sum, Average : INTEGER
count ← 1, index ← 1, sum ← 0, Average ← 0
INPUT N
FOR count 1 TO N
    FOR index 1 TO 5
        INPUT subject marks
        IF subject marks < 33 THEN
            OUTPUT "Fail - Subject Deficiency"
        ELSE
            sum ← sum + subject marks
        ENDIF
    NEXT index
    Average ← sum DIV 5
    IF Average ≥ 80 THEN
        OUTPUT "Distinction"
    ELSE
        IF Average ≥ 60 AND Average < 80 THEN
            OUTPUT "Pass"
        ELSE
            OUTPUT "Fail"
        ENDIF
    ENDIF
NEXT count
```